

Architecture Comparison Table

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How to read this table

Rows (Y-axis) - Data Package

Each row is one (target × split-ratio × split-strategy) combination. **target** = headache or migraine. **ratio** = train/val/test split sizes (e.g. 70_15_15 or 70_30 / 80_20 with chronological cal sub-split). **split** = how rows are assigned: *chrono* (date-percentile cuts), *stratified* (class-balanced random shuffle), *patient* (whole-patient holdout).

Columns (X-axis) - Architecture

Each column is one (model × version × feature-set) combination. Top line: model name. Middle line (if present): version. Bottom yellow tag [...]: feature-set abbreviation - see glossary below.

Cells

All 11 metrics on the locked test set. Each metric has its own colour scale (see *Colour legend*). The MCC row substitutes "Cal-Optimal" for "Optimal" automatically when the source is CV. A cell tagged **H+CV** has both hold-out and 5-fold CV results (cell shows hold-out). A cell tagged **CV** only has CV results - used as fallback when hold-out is missing. Empty (-) means no result file for that combination.

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15 chrono	70_15_15 patient
Addition 0	arch: blended xgb lr spano2026 fs: [spano]	-	H+CV	
			AUROC 0.534 [0.394-0.671]	-
			AUPRC 0.168 [0.089-0.264]	
			Brier 0.132 [0.097-0.171] ↓	
			ECE10 0.078 [0.038-0.126] ↓	
			CalSlope0.012 [-0.104-0.073]	
			MCC -0.012 [-0.116-0.181]	
			Sens≥0.50.262 [0.067-0.471]	
			Acc (base!) 0.816 [0.750-0.875]	
			Prec 0.124 [0.000-0.417]	
			Recall 0.050 [0.000-0.167]	
			F1 0.069 [0.000-0.235]	
Addition 0	arch: stacked 2xgb meta lr fs: [full]	-	H+CV	AUROC 0.639 [0.584-0.693]
			AUROC 0.507 [0.362-0.643]	AUPRC 0.376 [0.295-0.456]
			AUPRC 0.189 [0.094-0.317]	Brier 0.157 [0.143-0.173] ↓
			Brier 0.129 [0.096-0.163] ↓	ECE10 0.064 [0.036-0.091] ↓
			ECE10 0.076 [0.031-0.124] ↓	CalSlope1.448 [0.930-1.987]
			CalSlope0.190 [-0.831-1.228]	MCC 0.154 [0.075-0.236]
			MCC 0.016 [-0.129-0.206]	Sens≥0.50.453 [0.371-0.534]
			Sens≥0.50.417 [0.188-0.643]	Acc 0.649 [0.613-0.685] (base!)
			Acc (base!) 0.794 [0.728-0.860]	Prec 0.293 [0.235-0.355]
			Prec 0.157 [0.000-0.400]	Recall 0.489 [0.406-0.565]
			Recall 0.108 [0.000-0.263]	F1 0.366 [0.301-0.431]
			F1 0.124 [0.000-0.286]	

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15 chrono	70_15_15 patient
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: pareto_AUPRC_slope/auprc_max	-	-	-
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: pareto_AUPRC_slope/knee	-	-	-
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: pareto_AUPRC_slope/slope_closest	-	-	-
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: pareto_AUROC_slope/auroc_max	-	AUROC 0.405 [0.269-0.541] AUPRC 0.137 [0.075-0.223] Brier 0.144 [0.112-0.179] ↓ ECE10 0.140 [0.092-0.194] ↓ CalSlope -0.577 [-1.845-0.254] MCC -0.191 [-0.335--0.043] Sens≥0.5 0.262 [0.067-0.458]	AUROC 0.626 [0.574-0.675] AUPRC 0.324 [0.253-0.401] Brier 0.160 [0.147-0.176] ↓ ECE10 0.052 [0.027-0.079] ↓ CalSlope 1.029 [0.586-1.437] MCC 0.130 [0.056-0.208] Sens≥0.5 0.482 [0.403-0.565]

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: pareto_AUROC_slope/knee	-	chrono	patient
			Acc (base!) 0.434 [0.348-0.522]	Acc (base!) 0.605 [0.578-0.641]
			Prec 0.074 [0.016-0.136]	Prec 0.271 [0.220-0.328]
			Recall 0.262 [0.067-0.458]	Recall 0.533 [0.454-0.618]
			F1 0.114 [0.025-0.202]	F1 0.359 [0.301-0.423]
			AUROC 0.410 [0.276-0.549]	AUROC 0.630 [0.578-0.683]
			AUPRC 0.134 [0.076-0.212]	AUPRC 0.338 [0.266-0.414]
			Brier ↓ 0.141 [0.109-0.176]	Brier ↓ 0.159 [0.145-0.174]
			ECE10 ↓ 0.129 [0.078-0.187]	ECE10 ↓ 0.058 [0.032-0.086]
			CalSlope -0.766 [-1.058-0.257]	CalSlope 1.107 [0.677-1.543]
			MCC -0.106 [-0.276-0.065]	MCC 0.118 [0.041-0.199]
			Sens≥0.5 0.365 [0.143-0.588]	Sens≥0.5 0.360 [0.278-0.440]
			Acc (base!) 0.428 [0.345-0.507]	Acc (base!) 0.666 [0.632-0.704]
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: pareto_AUROC_slope/slope_closest	-	Prec 0.107 [0.044-0.182]	Prec 0.283 [0.222-0.351]
			Recall 0.418 [0.214-0.632]	Recall 0.396 [0.312-0.479]
			F1 0.169 [0.073-0.275]	F1 0.329 [0.263-0.397]
			AUROC 0.493 [0.342-0.643]	AUROC 0.640 [0.586-0.693]
			AUPRC 0.196 [0.095-0.349]	AUPRC 0.355 [0.273-0.436]
			Brier ↓ 0.131 [0.099-0.167]	Brier ↓ 0.158 [0.144-0.173]
			ECE10 ↓ 0.092 [0.045-0.140]	ECE10 ↓ 0.056 [0.030-0.083]
			CalSlope 0.001 [-1.108-1.110]	CalSlope 1.228 [0.788-1.675]
			MCC -0.004 [-0.174-0.158]	MCC 0.153 [0.071-0.227]
			Sens≥0.5 0.364 [0.160-0.583]	Sens≥0.5 0.431 [0.352-0.511]
			Acc (base!) 0.329 [0.259-0.404]	Acc (base!) 0.630 [0.592-0.668]
			Prec 0.139 [0.075-0.212]	Prec 0.287 [0.229-0.344]
			Recall 0.732 [0.526-0.933]	Recall 0.526 [0.441-0.606]
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: single_AUROC/HP020	-	F1 0.233 [0.133-0.331]	F1 0.371 [0.305-0.429]
			AUROC 0.503 [0.360-0.634]	AUROC 0.647 [0.595-0.697]
			AUPRC 0.160 [0.090-0.251]	AUPRC 0.364 [0.286-0.446]
			Brier ↓ 0.139 [0.107-0.173]	Brier ↓ 0.157 [0.143-0.172]
			ECE10 ↓ 0.117 [0.070-0.165]	ECE10 ↓ 0.062 [0.033-0.089]
			CalSlope -0.083 [-0.063-0.758]	CalSlope 1.435 [0.941-1.924]
			MCC -0.084 [-0.126--0.044]	MCC 0.144 [0.070-0.222]
			Sens≥0.5 0.471 [0.250-0.684]	Sens≥0.5 0.446 [0.369-0.533]
			Acc (base!) 0.816 [0.750-0.875]	Acc (base!) 0.651 [0.618-0.685]
			Prec 0.000 [0.000-0.000]	Prec 0.290 [0.232-0.357]
			Recall 0.000 [0.000-0.000]	Recall 0.468 [0.389-0.552]
			F1 0.000 [0.000-0.000]	F1 0.357 [0.297-0.422]
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: single_AUROC/HP050	-	AUROC 0.503 [0.360-0.634]	AUROC 0.621 [0.566-0.677]
			AUPRC 0.160 [0.090-0.251]	AUPRC 0.343 [0.269-0.424]
			Brier ↓ 0.139 [0.107-0.173]	Brier ↓ 0.160 [0.146-0.176]
			ECE10 ↓ 0.117 [0.070-0.165]	ECE10 ↓ 0.050 [0.025-0.075]

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
			chrono	patient
			CalSlope -0.083 [-0.062-0.753]	CalSlope 0.978 [0.575-1.368]
			MCC -0.084 [-0.126--0.044]	MCC 0.148 [0.069-0.225]
			Sens≥0.5 0.471 [0.250-0.684]	Sens≥0.5 0.388 [0.308-0.468]
			Acc (base!) 0.816 [0.750-0.875]	Acc (base!) 0.692 [0.635-0.728]
			Prec 0.000 [0.000-0.000]	Prec 0.309 [0.240-0.375]
			Recall 0.000 [0.000-0.000]	Recall 0.388 [0.308-0.468]
			F1 0.000 [0.000-0.000]	F1 0.343 [0.275-0.416]
			AUROC 0.487 [0.338-0.623]	AUROC 0.630 [0.575-0.684]
			AUPRC 0.164 [0.088-0.271]	AUPRC 0.352 [0.273-0.435]
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: single_AUROC/HP100	-	Brier 0.140 [0.106-0.176] ↓	Brier 0.160 [0.146-0.175] ↓
			ECE10 0.118 [0.070-0.169] ↓	ECE10 0.057 [0.033-0.083] ↓
			CalSlope -0.205 [-1.132-0.637]	CalSlope 0.983 [0.593-1.381]
			MCC 0.003 [-0.143-0.178]	MCC 0.218 [0.125-0.303]
			Sens≥0.5 0.262 [0.067-0.471]	Sens≥0.5 0.481 [0.400-0.563]
			Acc (base!) 0.787 [0.720-0.853]	Acc (base!) 0.755 [0.722-0.786]
			Prec 0.145 [0.000-0.364]	Prec 0.396 [0.309-0.486]
			Recall 0.104 [0.000-0.250]	Recall 0.345 [0.268-0.426]
			F1 0.118 [0.000-0.278]	F1 0.368 [0.291-0.446]
			AUROC 0.487 [0.338-0.623]	AUROC 0.630 [0.575-0.684]
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: single_AUROC/HP200	-	AUPRC 0.164 [0.088-0.271]	AUPRC 0.352 [0.273-0.435]
			Brier 0.140 [0.106-0.176] ↓	Brier 0.160 [0.146-0.175] ↓
			ECE10 0.118 [0.070-0.169] ↓	ECE10 0.057 [0.033-0.083] ↓
			CalSlope -0.205 [-1.132-0.637]	CalSlope 0.983 [0.593-1.381]
			MCC 0.003 [-0.143-0.178]	MCC 0.218 [0.125-0.303]
			Sens≥0.5 0.262 [0.067-0.471]	Sens≥0.5 0.481 [0.400-0.563]
			Acc (base!) 0.787 [0.720-0.853]	Acc (base!) 0.755 [0.722-0.786]
			Prec 0.145 [0.000-0.364]	Prec 0.396 [0.309-0.486]
			Recall 0.104 [0.000-0.250]	Recall 0.345 [0.268-0.426]
			F1 0.118 [0.000-0.278]	F1 0.368 [0.291-0.446]
Addition 0	arch: stacked 2xgb meta lr fs: [full] HP: single_AUROC/HP500	-	AUROC 0.487 [0.341-0.624]	AUROC 0.630 [0.575-0.684]
			AUPRC 0.155 [0.086-0.245]	AUPRC 0.352 [0.273-0.435]
			Brier 0.139 [0.108-0.175] ↓	Brier 0.160 [0.146-0.175] ↓
			ECE10 0.107 [0.058-0.159] ↓	ECE10 0.057 [0.033-0.083] ↓
			CalSlope -0.197 [-1.087-0.625]	CalSlope 0.983 [0.593-1.381]
			MCC -0.088 [-0.182-0.053]	MCC 0.218 [0.125-0.303]
			Sens≥0.5 0.365 [0.154-0.579]	Sens≥0.5 0.481 [0.400-0.563]
			Acc (base!) 0.750 [0.676-0.824]	Acc (base!) 0.755 [0.722-0.786]
			Prec 0.058 [0.000-0.200]	Prec 0.396 [0.309-0.486]
			Recall 0.050 [0.000-0.167]	Recall 0.345 [0.268-0.426]
			F1 0.053 [0.000-0.174]	F1 0.368 [0.291-0.446]

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
Addition 0	arch: stacked 2xgb meta lr fs: [no_rolling]	-	chrono	patient
			H+CV	AUROC 0.514 [0.462-0.564]
			AUROC 0.518 [0.373-0.670]	AUPRC 0.209 [0.174-0.247]
			AUPRC 0.189 [0.098-0.326]	Brier 0.169 [0.155-0.183]
			Brier ↓ 0.126 [0.095-0.160]	ECE10 0.058 [0.028-0.086]
			ECE10 ↓ 0.076 [0.026-0.127]	CalSlope -0.356 [-1.081-1.012]
			CalSlope -0.032 [-4.882-3.118]	MCC -0.092 [-0.124--0.047]
			MCC 0.015 [-0.150-0.200]	Sens≥0.5 0.834 [0.771-0.896]
			Sens≥0.5 0.584 [0.357-0.824]	Acc (base!) 0.751 [0.717-0.781]
			Acc (base!) 0.685 [0.618-0.750]	Prec 0.033 [0.000-0.127]
			Prec 0.149 [0.036-0.281]	Recall 0.007 [0.000-0.025]
			Recall 0.265 [0.071-0.474]	F1 0.012 [0.000-0.025]
			F1 0.188 [0.047-0.333]	
Addition 0	arch: stacked 2xgb meta lr fs: [park]	-	-	-
Addition 0	arch: stacked 2xgb meta lr fs: [spano]	-	H+CV	AUROC 0.620 [0.567-0.677]
			AUROC 0.535 [0.390-0.673]	AUPRC 0.376 [0.295-0.456]
			AUPRC 0.226 [0.104-0.394]	Brier 0.158 [0.144-0.174]
			Brier ↓ 0.127 [0.093-0.164]	ECE10 0.061 [0.031-0.089]
			ECE10 ↓ 0.078 [0.036-0.124]	CalSlope 1.362 [0.864-1.906]
			CalSlope 0.263 [-0.748-1.294]	MCC 0.069 [-0.006-0.146]
			MCC 0.059 [-0.083-0.306]	Sens≥0.5 0.424 [0.345-0.505]
			Sens≥0.5 0.313 [0.111-0.526]	Acc (base!) 0.407 [0.371-0.443]
			Acc (base!) 0.846 [0.787-0.897]	Prec 0.226 [0.188-0.265]
			Prec 0.265 [0.000-1.000]	Recall 0.764 [0.692-0.837]
			Recall 0.053 [0.000-0.176]	F1 0.348 [0.300-0.395]
			F1 0.086 [0.000-0.273]	
Addition 1	arch: tabpfn ver: v2-5-auto fs: [full]	-	-	-

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
Addition 1	arch: tabPFN ver: v2-5-real fs: [full]	-	chrono	patient
			H+CV	AUROC 0.644 [0.591-0.697]
			AUROC 0.519 [0.365-0.671]	AUPRC 0.379 [0.299-0.465]
			AUPRC 0.174 [0.094-0.277]	Brier 0.153 [0.136-0.172]
			Brier 0.126 [0.090-0.164] ↓	ECE10 0.044 [0.021-0.071] ↓
			ECE10 0.063 [0.026-0.110] ↓	CalSlope 0.895 [0.600-1.200]
			CalSlope -0.036 [-1.618-1.546]	MCC 0.131 [0.051-0.213]
			MCC 0.000 [0.000-0.000]	Sens≥0.5 0.417 [0.336-0.497]
			Sens≥0.5 0.158 [0.000-0.313]	Acc 0.663 [0.635-0.690] (base!)
			Acc 0.860 [0.801-0.912] (base!)	Prec 0.288 [0.225-0.352]
			Prec 0.000 [0.000-0.000]	Recall 0.424 [0.343-0.507]
			Recall 0.000 [0.000-0.000]	F1 0.342 [0.276-0.411]
			F1 0.000 [0.000-0.000]	
Addition 1	arch: tabPFN ver: v2-5-real fs: [no_rolling]	-	H+CV	AUROC 0.473 [0.418-0.526]
			AUROC 0.548 [0.405-0.691]	AUPRC 0.213 [0.171-0.263]
			AUPRC 0.186 [0.103-0.304]	Brier 0.169 [0.153-0.188] ↓
			Brier 0.131 [0.100-0.165] ↓	ECE10 0.057 [0.030-0.087] ↓
			ECE10 0.089 [0.043-0.142] ↓	CalSlope -0.093 [-0.591-0.403]
			CalSlope -0.118 [-2.231-1.993]	MCC -0.045 [-0.118-0.027]
			MCC 0.069 [-0.106-0.244]	Sens≥0.5 0.670 [0.594-0.747]
			Sens≥0.5 0.425 [0.200-0.667]	Acc 0.623 [0.585-0.661] (base!)
			Acc 0.635 [0.501-0.721] (base!)	Prec 0.177 [0.123-0.236]
			Prec 0.173 [0.075-0.286]	Recall 0.224 [0.158-0.292]
			Recall 0.425 [0.200-0.667]	F1 0.197 [0.140-0.256]
			F1 0.243 [0.105-0.371]	
Addition 1	arch: tabPFN ver: v2-5-real fs: [park]	-		
Addition 1	arch: tabPFN ver: v2-6 fs: [full]	-	H+CV	AUROC 0.642 [0.588-0.696]
			AUROC 0.518 [0.354-0.673]	AUPRC 0.388 [0.306-0.471]
			AUPRC 0.177 [0.096-0.283]	Brier 0.153 [0.135-0.172] ↓
			Brier 0.126 [0.091-0.164] ↓	ECE10 0.046 [0.023-0.074] ↓
			ECE10 0.077 [0.034-0.126] ↓	CalSlope 0.866 [0.589-1.156]
			CalSlope -0.024 [-1.495-1.447]	

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
			chrono	patient
			MCC -0.083 [-0.124--0.045]	MCC 0.144 [0.064-0.222]
			Sens≥0.5 0.208 [0.000-0.400]	Sens≥0.5 0.453 [0.370-0.534]
			Acc (base!) 0.817 [0.750-0.875]	Acc (base!) 0.633 [0.500-0.670]
			Prec 0.000 [0.000-0.000]	Prec 0.284 [0.228-0.343]
			Recall 0.000 [0.000-0.000]	Recall 0.504 [0.422-0.581]
			F1 0.000 [0.000-0.000]	F1 0.363 [0.301-0.427]
Addition 1	arch: tabPFN ver: v2-6 fs: [no_rolling]	-	H+CV	AUROC 0.469 [0.419-0.521]
			AUROC 0.489 [0.331-0.640]	AUPRC 0.207 [0.166-0.253]
			AUPRC 0.167 [0.091-0.269]	Brier 0.170 [0.154-0.189]
			Brier 0.132 [0.101-0.165] ↓	ECE10 0.052 [0.027-0.080] ↓
			ECE10 0.090 [0.037-0.140] ↓	Calslope -0.106 [-0.054-0.421]
			Calslope -0.144 [-2.246-1.341]	MCC -0.022 [-0.091-0.052]
			MCC 0.044 [-0.137-0.208]	Sens≥0.5 0.490 [0.414-0.573]
			Sens≥0.5 0.685 [0.461-0.883]	Acc (base!) 0.661 [0.620-0.693]
			Acc (base!) 0.420 [0.330-0.507]	Prec 0.191 [0.130-0.266]
			Prec 0.152 [0.079-0.233]	Recall 0.195 [0.133-0.266]
			Recall 0.685 [0.461-0.883]	F1 0.192 [0.132-0.255]
			F1 0.247 [0.139-0.359]	
Addition 1	arch: tabPFN ver: v2-6 fs: [park]	-	-	-
			-	-
			-	-
			-	-
			-	-
			-	-
			-	-
			-	-
			-	-
			-	-
			-	-
			-	-
			-	-
			-	-
Addition 1	arch: tabPFN ver: v3-binary fs: [full]	-	H+CV	AUROC 0.647 [0.593-0.706]
			AUROC 0.510 [0.363-0.657]	AUPRC 0.387 [0.307-0.471]
			AUPRC 0.170 [0.095-0.274]	Brier 0.152 [0.135-0.171] ↓
			Brier 0.126 [0.091-0.164] ↓	ECE10 0.033 [0.014-0.058] ↓
			ECE10 0.074 [0.035-0.124] ↓	Calslope 0.923 [0.641-1.231]
			Calslope -0.118 [-1.806-1.192]	MCC 0.128 [0.061-0.185]
			MCC -0.056 [-0.095-0.000]	Sens≥0.5 0.417 [0.336-0.497]
			Sens≥0.5 0.158 [0.000-0.333]	Acc (base!) 0.415 [0.376-0.451]
			Acc (base!) 0.838 [0.772-0.897]	Prec 0.240 [0.202-0.279]
			Prec 0.000 [0.000-0.000]	Recall 0.835 [0.774-0.896]
			Recall 0.000 [0.000-0.000]	F1 0.372 [0.324-0.419]
			F1 0.000 [0.000-0.000]	

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
Addition 1	arch: tabPFN ver: v3-binary fs: [no_rolling]	-	chrono	patient
			H+CV	AUROC 0.473 [0.418-0.536]
			AUROC 0.537 [0.399-0.674]	AUPRC 0.211 [0.170-0.266]
			AUPRC 0.175 [0.099-0.276]	Brier 0.170 [0.154-0.188]
			Brier 0.132 [0.101-0.165] ↓	ECE10 0.062 [0.035-0.093] ↓
			ECE10 0.088 [0.037-0.139] ↓	CalSlope -0.123 [-0.718-0.498]
			CalSlope -0.251 [-2.694-1.481]	MCC -0.019 [-0.090-0.055]
			MCC 0.009 [-0.155-0.192]	Sens≥0.5 0.504 [0.422-0.587]
			Sens≥0.5 0.844 [0.667-1.000]	Acc 0.654 [0.619-0.689] (base!)
			Acc 0.679 [0.603-0.755] (base!)	Prec 0.193 [0.132-0.258]
			Prec 0.146 [0.034-0.273]	Recall 0.210 [0.146-0.278]
			Recall 0.265 [0.071-0.474]	F1 0.201 [0.141-0.265]
			F1 0.185 [0.045-0.327]	
Addition 1	arch: tabPFN ver: v3-binary fs: [park]	-	-	-
Addition 1	arch: tabPFN ver: v3-default fs: [full]	-	H+CV	AUROC 0.646 [0.594-0.706]
			AUROC 0.515 [0.365-0.666]	AUPRC 0.391 [0.309-0.474]
			AUPRC 0.174 [0.096-0.281]	Brier 0.152 [0.135-0.171] ↓
			Brier 0.126 [0.091-0.164] ↓	ECE10 0.036 [0.015-0.062] ↓
			ECE10 0.077 [0.034-0.127] ↓	CalSlope 0.901 [0.619-1.201]
			CalSlope -0.051 [-1.438-1.324]	MCC 0.145 [0.067-0.226]
			MCC -0.056 [-0.095-0.000]	Sens≥0.5 0.410 [0.329-0.493]
			Sens≥0.5 0.158 [0.000-0.315]	Acc 0.638 [0.603-0.674] (base!)
			Acc 0.838 [0.772-0.897] (base!)	Prec 0.286 [0.231-0.347]
			Prec 0.000 [0.000-0.000]	Recall 0.497 [0.413-0.579]
			Recall 0.000 [0.000-0.000]	F1 0.362 [0.301-0.429]
			F1 0.000 [0.000-0.000]	
Addition 1	arch: tabPFN ver: v3-default fs: [no_rolling]	-	H+CV	AUROC 0.469 [0.417-0.519]
			AUROC 0.510 [0.374-0.654]	AUPRC 0.209 [0.168-0.257]
			AUPRC 0.167 [0.092-0.263]	Brier 0.170 [0.154-0.189] ↓
			Brier 0.131 [0.100-0.165] ↓	ECE10 0.062 [0.035-0.090] ↓
			ECE10 0.085 [0.039-0.136] ↓	CalSlope -0.111 [-0.682-0.420]
			CalSlope -0.357 [-2.631-1.389]	

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
			chrono	patient
			MCC -0.024 [-0.183-0.151]	MCC -0.021 [-0.091-0.053]
			Sens≥0.5 0.844 [0.667-1.000]	Sens≥0.5 0.568 [0.488-0.646]
			Acc (base!) 0.642 [0.580-0.720]	Acc (base!) 0.653 [0.610-0.696]
			Prec 0.127 [0.029-0.235]	Prec 0.192 [0.133-0.257]
			Recall 0.265 [0.071-0.474]	Recall 0.210 [0.146-0.276]
			F1 0.170 [0.041-0.300]	F1 0.200 [0.141-0.265]
Addition 1	arch: tabpfn ver: v3-default fs: [park]	-	-	-
Addition 2	arch: ? fs: [?]	CV	-	-
		AUROC		
		AUPRC		
		Brier ↓		
		ECE10 ↓		
		CalSlope		
		MCC		
		Sens≥0.5		
		Acc (base!)		
		Prec		
		Recall		
		F1		
Addition 5	arch: add0 stacked fs: [full]	-	-	-
Addition 5	arch: add1 tabpfn fs: [full]	-	-	-

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
			chrono	patient
Addition 5	arch: add4 window mlp fs: [full]	-	-	-
Addition 5	arch: partial pool fs: [no_rolling]	-	-	-
Addition 5	arch: partial pool fs: [park]	-	-	-

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15	70_15_15
			chrono	patient
Addition 5	arch: per patient fs: [no_rolling]	-	-	-
Addition 5	arch: per patient fs: [park]	-	-	-
Addition 5	arch: pooled fs: [no_rolling]	-	-	-
Addition 5	arch: pooled fs: [park]	-	-	-

rows ↓ Architecture cols → Data Package		?		
		?	70_15_15 chrono	70_15_15 patient
Addition 5				
	arch: pooled lr fs: [full]	-	-	-

Best-cell markers (strict CI separation)

Marker	Meaning
 left edge, dark green	Row-best: for this architecture, this data-package's 95% CI on this metric does not overlap any other data-package's CI in the row. Statistically distinguishable best across data packages.
 upper edge, deep violet	Column-best: for this data-package, this architecture's 95% CI on this metric does not overlap any other architecture's CI in the column. Statistically distinguishable best across architectures.
 both edges	Cell wins both directions independently. Rare on small data: not only is this architecture the clear winner on this data-package, but this data-package is also the clear winner for this architecture.
no edge	No statistically distinguishable winner in this row/column for this metric at our sample size. The default reading; absence of marker is itself an honest scientific signal.

Colour legend (per-metric scale)

Metric	Direction	Reference range	Scale
AUROC	pivot = 0.5 (random); negative = anti-predictive	0.3 ... 0.85	 → 0.5 → 
AUPRC	↑ higher = better	0.05 ... 0.6	 →  →  Comparing AUPRC across the headache (~25% positive) and migraine (~5% positive) columns is misleading because the baseline differs; compare within a target only.
Brier ↓	↓ lower = better	0.0 ... 0.25	 →  → 

ECE10 ↓	↓ lower = better	0.0 ... 0.15	best → mid → worst
CalSlope	target = 1.0 (deviation in either direction is worse)	val - 1.0 up to 1.0	[2.0] → 1.0 → [0.0] Logistic-recalibration slope. 1.0 = perfect calibration; <1.0 = predictions are too extreme (the classical overfitting fingerprint); >1.0 = predictions are too conservative. Required by TRIPOD+AI alongside ECE/Brier.
MCC	pivot = 0.0 (random); negative = anti-predictive	-0.3 ... 0.5	anti → 0.0 → good
Sens≥0.5	↑ higher = better	0.2 ... 1.0	worst → mid → best
Acc (base!)	↑ higher = better	0.7 ... 1.0	worst → mid → best Base-rate-dominated on imbalanced targets. On migraine (~5% positive rate) a constant-negative predictor reaches ~95% accuracy, so this column should not be read as a headline. MCC, AUPRC and Brier are the primary metrics.
Prec	↑ higher = better	0.0 ... 0.6	worst → mid → best
Recall	↑ higher = better	0.0 ... 0.8	worst → mid → best
F1	↑ higher = better	0.0 ... 0.6	worst → mid → best

Feature-set glossary

Tag	Meaning
[?]	(no description)
[full]	all engineered features (today's triggers + rolling/lag/interaction)
[no_rolling]	today's triggers only - no temporal aggregation, lag, or streaks
[park]	Park (2016) [Tab. 4] stepwise-selected triggers: stress, hormonal_changes, noise, alcohol, overeating, travel (migraine target only)
[spano]	Spano (2026) feature subset - matches the prior-work replication

Feature-set inclusion (region counts)

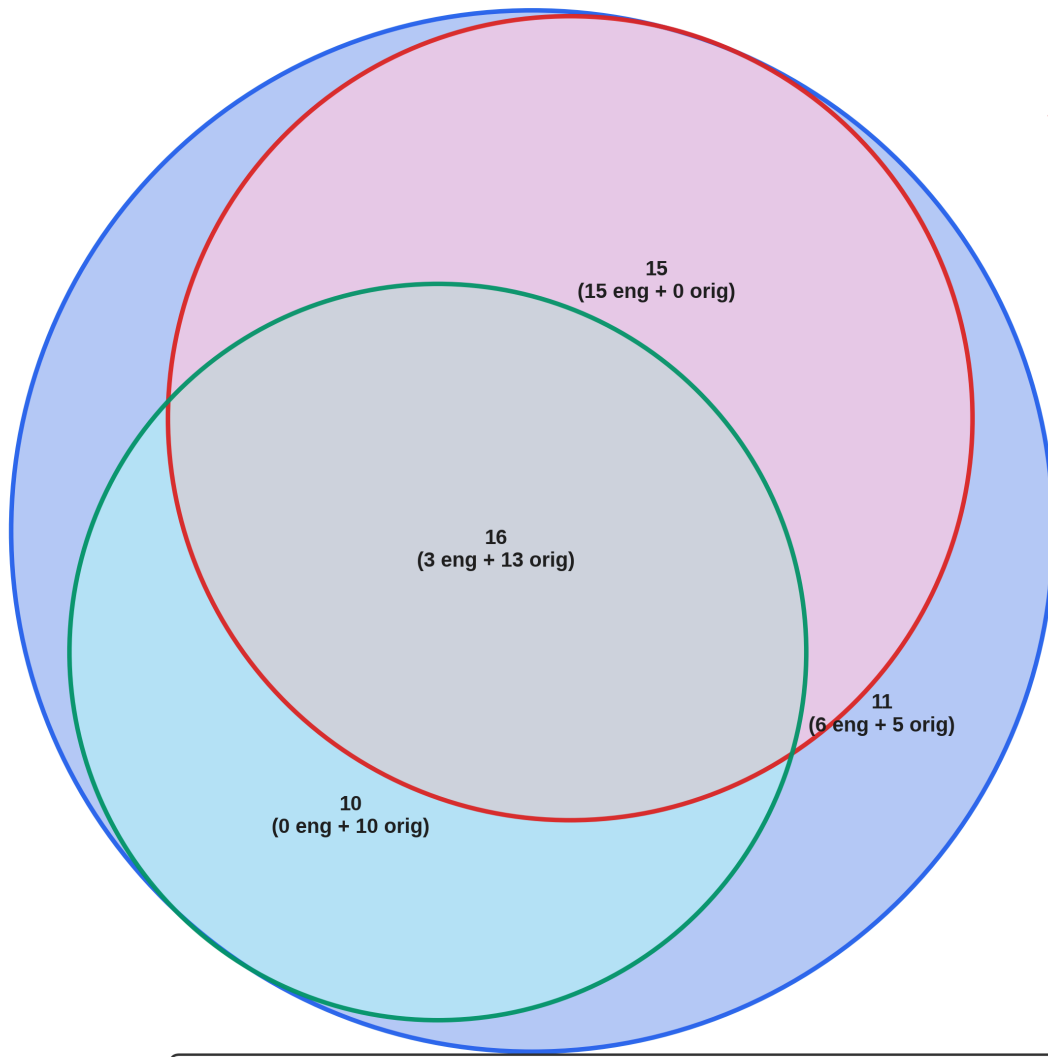
Computed from the canonical `diary_train.parquet` by running both filters (`remove_non_spano_features`, `remove_rolling_features`) at aggregation time. **full** = all engineered columns; **spano** = Spano-faithful subset; **no_rolling** = drops rolling/lag/interaction features. Each region label shows *total (engineered + original SHD columns)*.

Feature-set inclusion - region counts (engineered + original SHD columns)

eng = engineered (rolling / lag / interaction / state-derived)
orig = original SHD column or 1:1 rename

full (52)

spano (31)



Park (2016) stepwise-selected: 6 features
5 subset full: alcohol_today, noise_today, overeating_today, stress_today, travel_today
1 derived (not in full): hormonal_changes_today

no_rolling (26)

Feature-set inclusion (every feature by name)

Same three sets as above, but each region lists the actual feature names. • **green** = original SHD column or 1:1 rename. • **purple bold** = engineered (rolling, lag, interaction, or state-derived).

Feature-set structure (region counts)

full (52)

spano (31)

no rolling (26)

Feature-set structure (region counts)

full n spano n no_rolling (16)

- alcohol_today
- any_sleep_issue_today
- cheese_chocolate_today
- daw
- excessive_caffeine_today
- excessive_smoking_today
- exercise_today
- irregular_meals_today
- lack_of_sleep_today
- menstruation_today
- overeating_today
- oversleeping_today
- ovulation_today
- stress_today
- travel_today
- weather_change_today

full n no_rolling (not spano) (10)

- emotional_changes_today
- exercise_as_trigger_today
- inappropriate_lighting_today
- moderate_exercise_min
- no_exercise_today
- noise_today
- physical_fatigue_today
- specific_smells_today
- sunlight_today
- vigorous_exercise_min

spano only (11)

- cheese_chocolate
- days_since_last_record
- excessive_smoking
- exercise_as_trigger
- headache_free_streak
- inappropriate_lighting
- recording_gap_flag
- sleep_disruption_today
- sunlight
- weather_change_yesterday
- weather_headache_interaction

full n spano (not no_rolling) (15)

- consecutive_exercise_days
- consecutive_sedentary_days
- consecutive_stress_days
- consecutive_trigger_days
- consecutive_weather_changes
- days_since_last_migraine
- exercise_days_7day
- migraine_rate_last3
- migraine_rate_last7
- migraine_yesterday
- recent_weekend_sleep_issues
- sleep_debt_3day
- sleep_variability_7day
- stress_drop_today
- weather_instability_3day

Park (2016) [Tab. 4] subset (6)

- alcohol_today
- hormonal_changes_today
- noise_today
- overeating_today
- stress_today
- travel_today

● original SHD column or 1:1 rename ● engineered: rolling / lag / interaction / state-derived

experiment/<addition>/<target>/<feature_set>/<architecture>/[<version>]/<datasplit>/<splittype>/[<hyperparameter>]. Each column has its own colour; node labels use the directory name verbatim.

Discovered experiment leaves (470 total) - icicle plot [Shneiderman 1992, Andrews 2019]: row = path level, cell width is proportional to descendant-leaf count, children align under parent

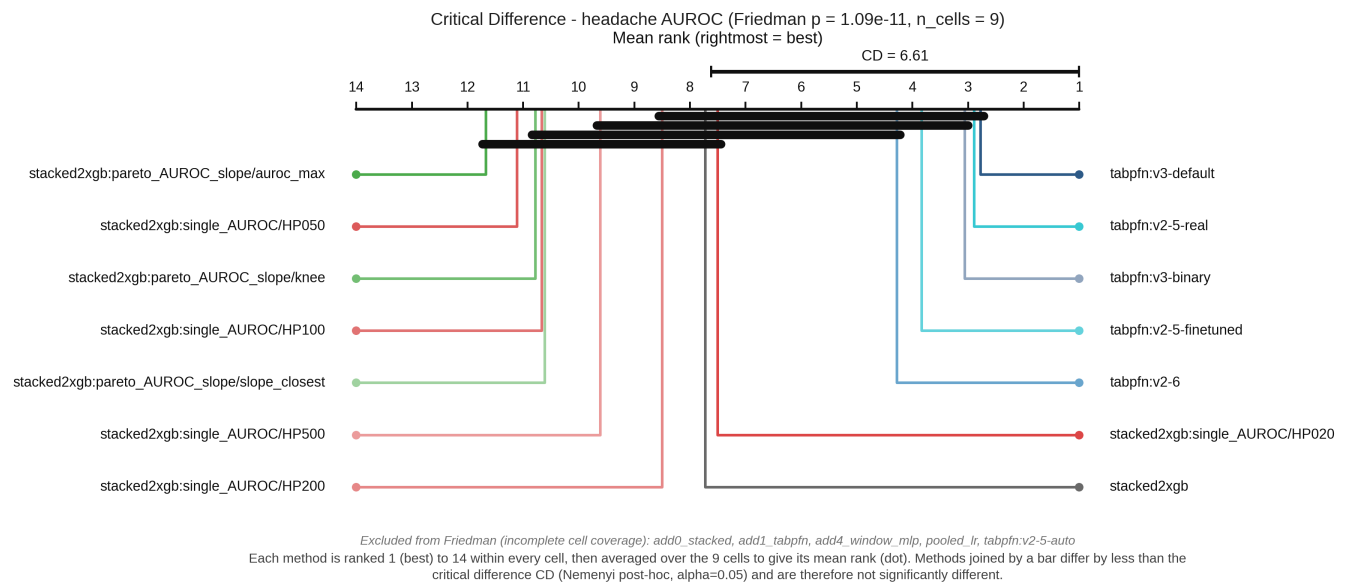


- **Critical Difference (CD)** [Demsar 2006, JMLR] - Friedman omnibus + Nemenyi post-hoc on per-cell ranks. Architectures connected by a heavy horizontal bar are not significantly different at $\alpha=0.05$. The "CD = X" bar at the top is the minimum mean-rank gap required for significance.
- **Performance profile** [Dolan & Moré 2002, Math. Prog.] - cumulative fraction of cells where each architecture is within factor τ of the best architecture in that cell. Curves that rise quickly (top-left) dominate; flatter curves are unstable.
- **Rank slopegraph** [Tufté 2001] - per-architecture rank trajectory across (ratio, split) cells. Stable architectures are flat; unstable architectures crisscross.

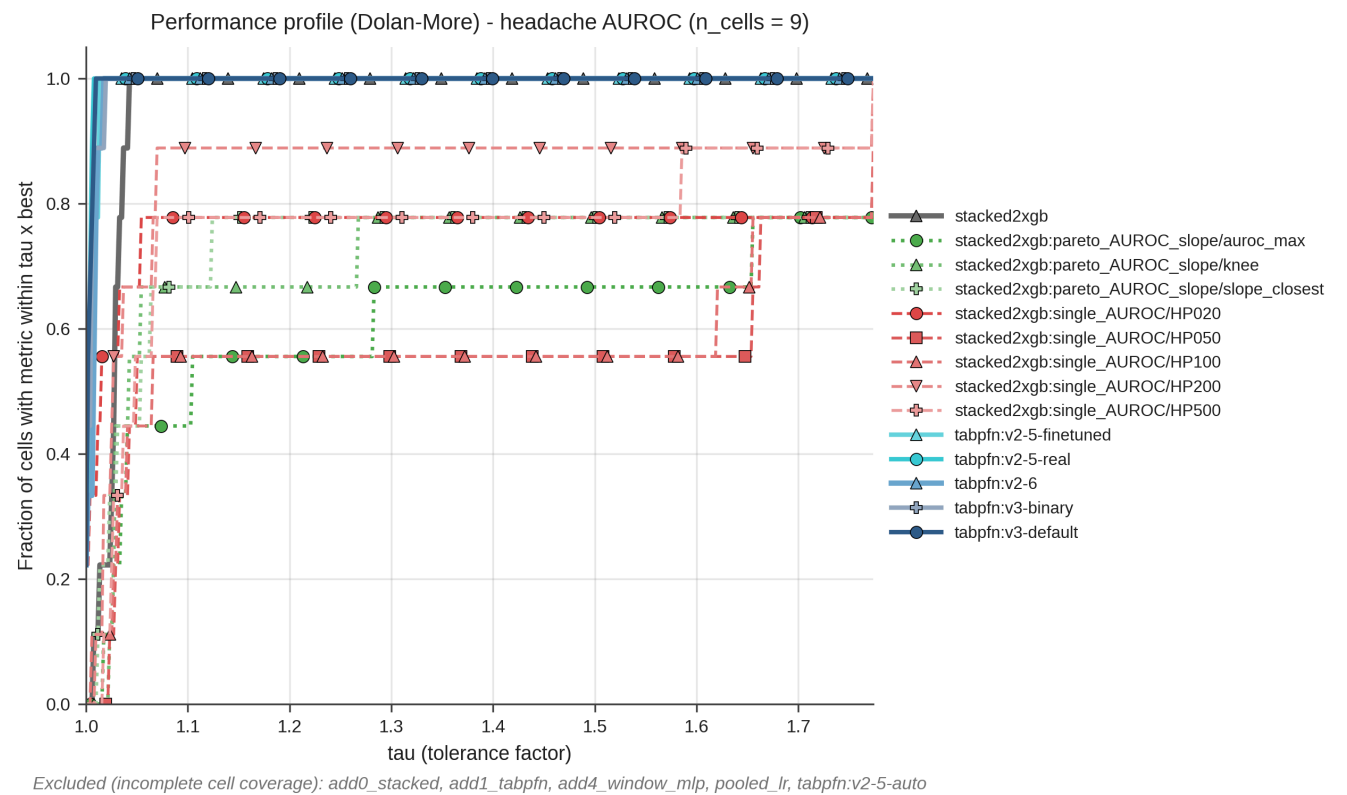
Grid restricted to full_features so every cell carries the same set of competing architectures (Friedman matched-design requirement).

Headache - AUROC across full_features cells

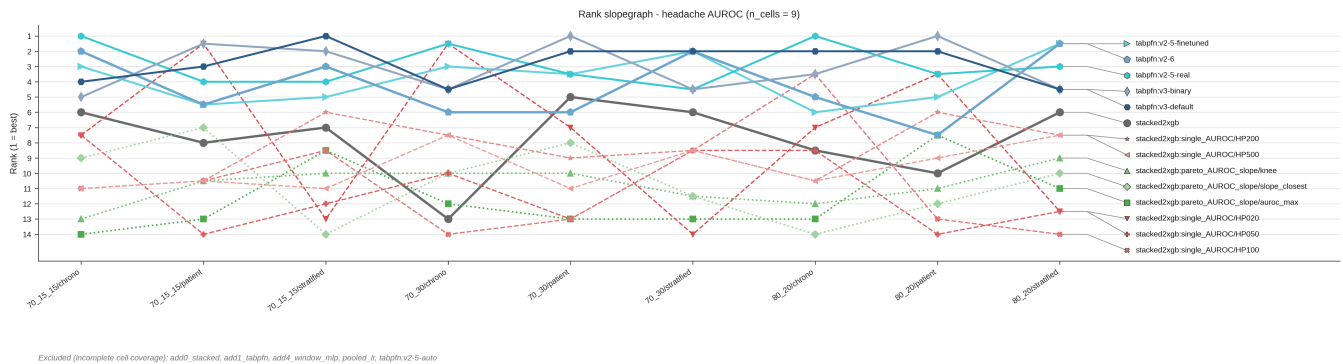
Critical Difference (Friedman + Nemenyi)



Performance profile (Dolan-More)

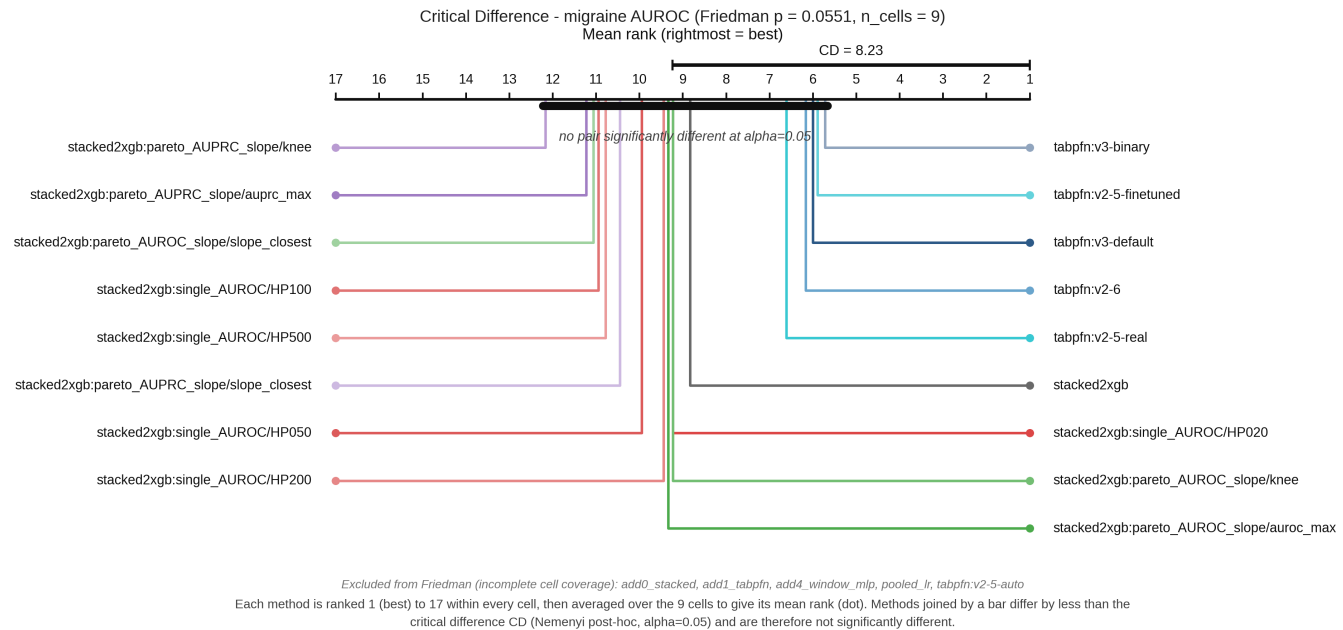


Rank slopegraph across (ratio, split)



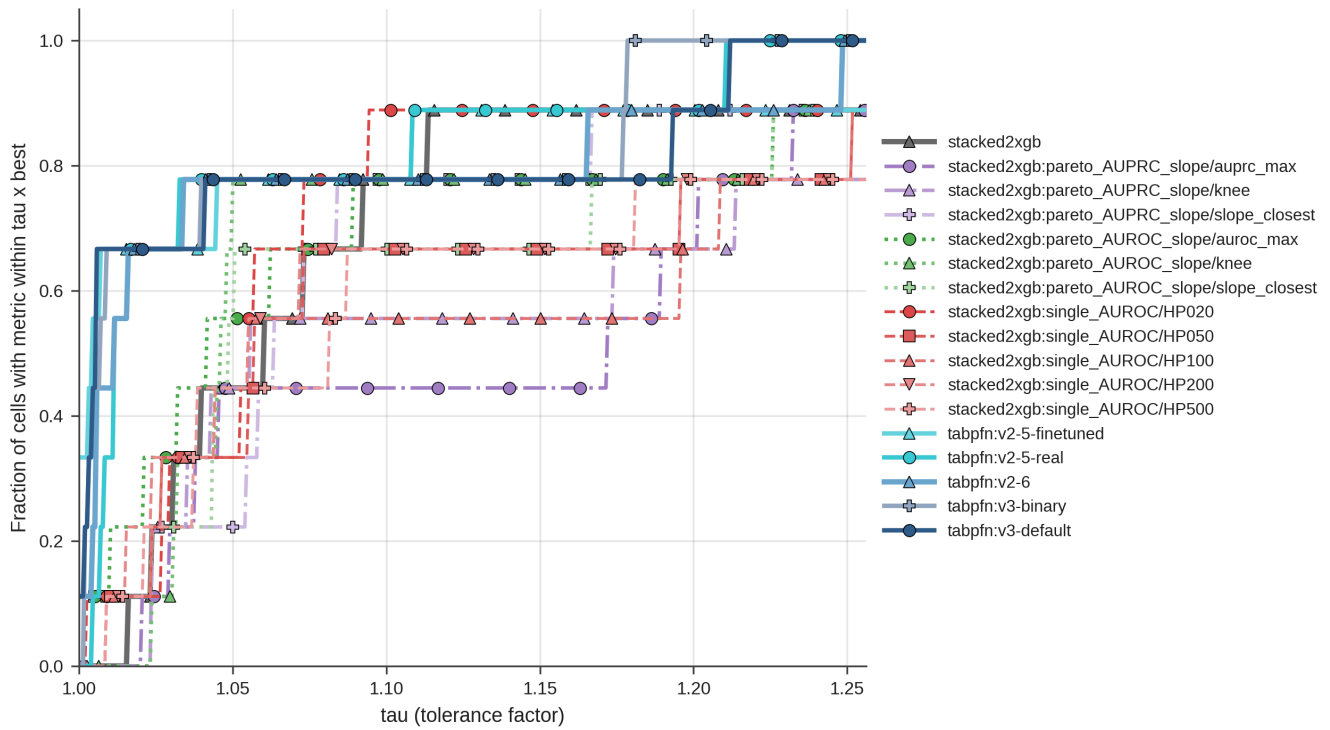
Migraine - AUROC across full_features cells

Critical Difference (Friedman + Nemenyi)

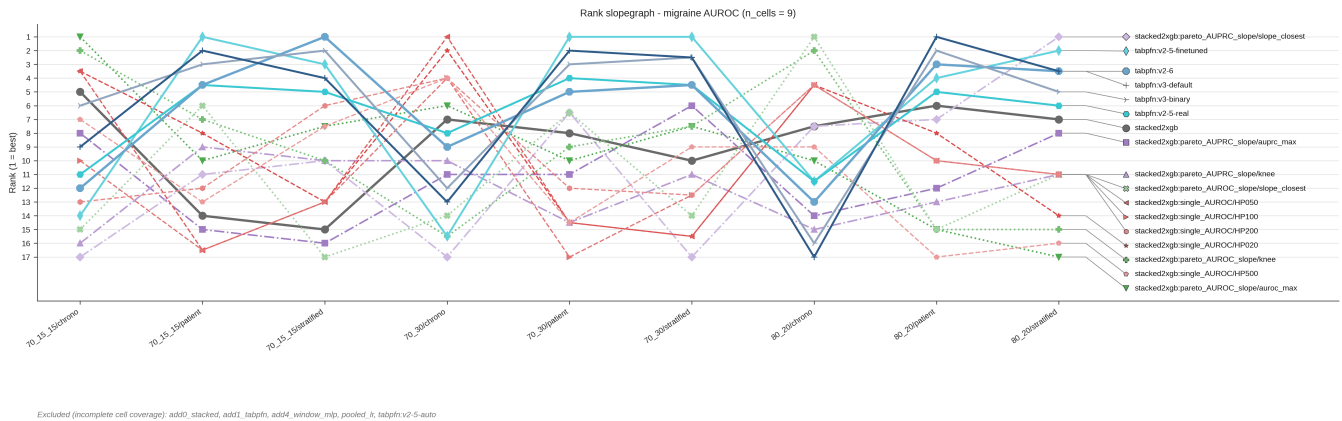


Performance profile (Dolan-More)

Performance profile (Dolan-More) - migraine AUROC (n_cells = 9)



Rank slopegraph across (ratio, split)



Why `blended_xgb_lr_spano2026` appears only at the canonical 70_15_15 / chrono cell

The architecture's held-out calibration set drives **four sequential optimisation steps**: per-base isotonic + Platt calibrators, alpha grid search, final-calibrator selection, and downstream operating-threshold selection. Because the isotonic calibrators effectively memorise the calibration set, threshold-derived metrics (MCC, Sensitivity ≥ 0.5 , F1) on test are unreliable; AUROC and AUPRC remain trustworthy because they are rank-based and calibration-invariant.

The architecture is preserved as a faithful replication of the prior bachelor-thesis baseline (Spano 2026, single operating point). Fanning it out across ratios and split types would add cells whose threshold metrics could not be cleanly compared. The methodologically clean comparator is `stacked2xgb_meta_lr`, which uses two-parameter Platt calibration only and is fanned out to the full grid.